Advanced Object Oriented Programming

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高级面向对象编程

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01

High-level and low-level language

01

高级语言和低级语言

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High-level and low-level language

- Instructions that the CPU can recognize are machine language
  - If 5 volts is applied to the CPU's contacts, it is 1, otherwise it is 0.
  - Expressed in **binary** or converted to **hexadecimal**
- High-level programming language is a definition of machine language that is dependent on the device as a string representation that is easy for humans to understand.
  - **C, C++**
  - C#
  - **Java**
  - Python

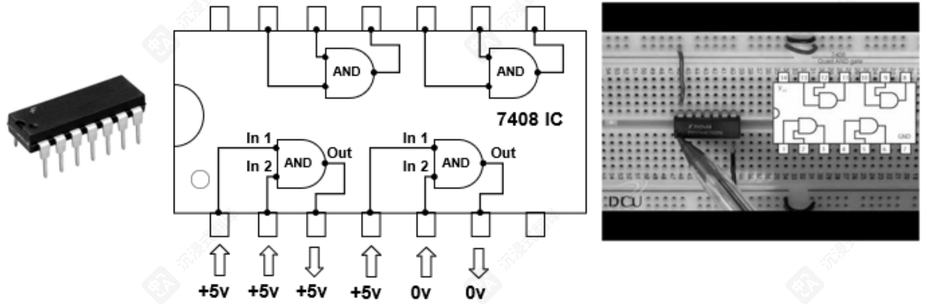
高级语言和低级语言

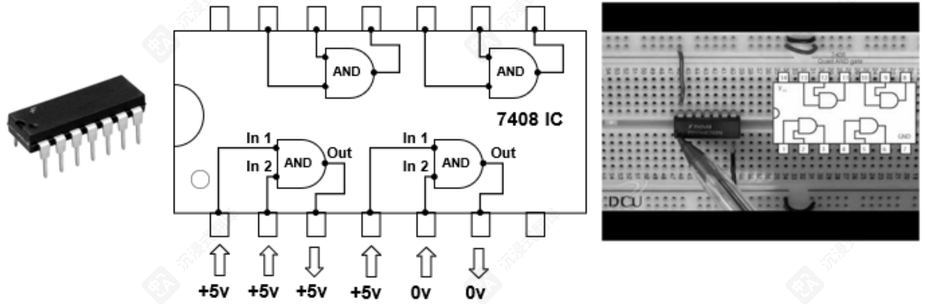
- CPU能识别的指令是机器语言
  - 如果5伏特施加到CPU的触点上，就是1，否则就是0。
  - 用二进制表示或转换为十六进制
- 高级编程语言 is a 定义 of 机器
  一种依赖于设备作为人类易于理解的字符串表示形式的语言。

  - **C, C++**
  - C#
  - **Java**
  - Python

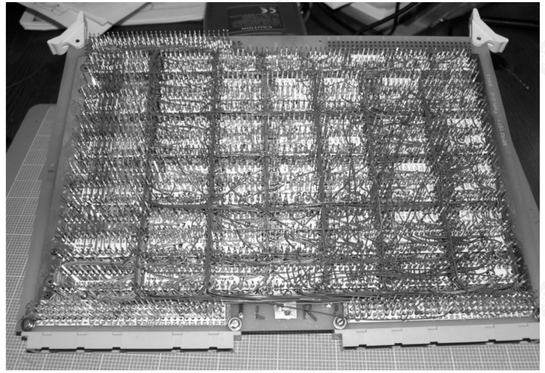
Digital Circuit

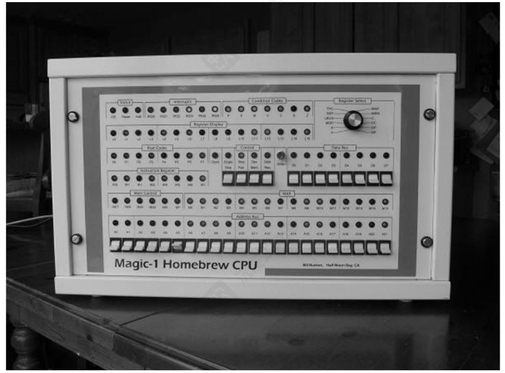
数字电路





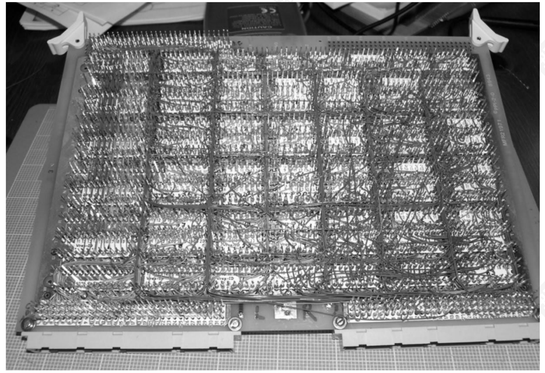
Homebrew CPU

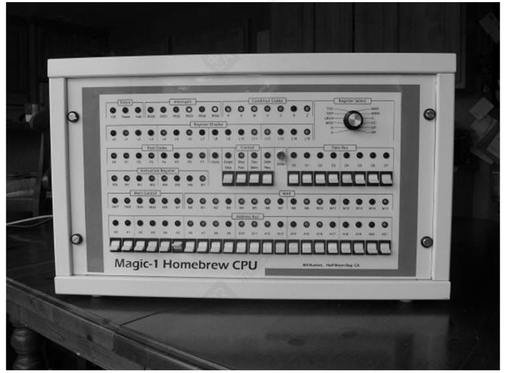




http://www.homebrewcpu.com/

Homebrew CPU





http://www.homebrewcpu.com/

Machine Code

```
메모리 1
주소: 0x00007FF65C4F1AC0   열: 8
0x00007FF65C4F1AC0  48 89 4c 24 08 55 57 48  H?L$.UWH
0x00007FF65C4F1AC8  81 ec 08 01 00 00 48 8d  ??....H?
0x00007FF65C4F1AD0  6c 24 20 48 8d 0d 36 05  l$ H?.6.
0x00007FF65C4F1AD8  01 00 e8 b4 f8 ff ff b9  ..???..?
0x00007FF65C4F1AE0  48 00 00 00 ff 15 1e f8  H......?
0x00007FF65C4F1AE8  00 00 48 89 45 08 41 b8  ..H?E.A?
0x00007FF65C4F1AF0  48 00 00 00 33 d2 48 8b  H...3?H?
0x00007FF65C4F1AF8  4d 08 e8 e0 f7 ff ff 48  M.???..H
0x00007FF65C4F1B00  8b 45 08 4c 8b 85 00 01  ?E.L??..
0x00007FF65C4F1B08  00 00 ba 40 00 00 00 48  ..?@...H
0x00007FF65C4F1B10  8b c8 ff 15 08 f8 00 00  ??...?..
```

```
mov         qword ptr [rsp+8],rcx
push        rbp
push        rdi
sub         rsp,108h
lea         rbp,[rsp+20h]
lea         rcx,[__AA290256_SingleList@c (
call        __CheckForDebuggerJustMyCode (
```

- Machine code is a combination of electrical signals (0, 1).
- It is expressed in **hexadecimal** rather than binary.
- **0x** indicates hexadecimal.
- It is usually expressed in **disassembly** code.

机器码

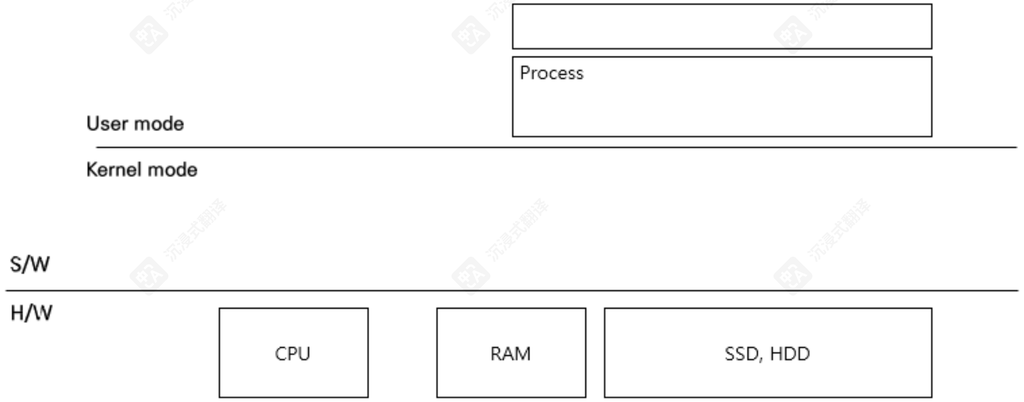

```
메모리 1
주소: 0x00007FF65C4F1AC0   열: 8
0x00007FF65C4F1AC0  48 89 4c 24 08 55 57 48  H?L$.UWH
0x00007FF65C4F1AC8  81 ec 08 01 00 00 48 8d  ??....H?
0x00007FF65C4F1AD0  6c 24 20 48 8d 0d 36 05  l$ H?.6.
0x00007FF65C4F1AD8  01 00 e8 b4 f8 ff ff b9  ..???..?
0x00007FF65C4F1AE0  48 00 00 00 ff 15 1e f8  H......?
0x00007FF65C4F1AE8  00 00 48 89 45 08 41 b8  ..H?E.A?
0x00007FF65C4F1AF0  48 00 00 00 33 d2 48 8b  H...3?H?
0x00007FF65C4F1AF8  4d 08 e8 e0 f7 ff ff 48  M.???..H
0x00007FF65C4F1B00  8b 45 08 4c 8b 85 00 01  ?E.L??..
0x00007FF65C4F1B08  00 00 ba 40 00 00 00 48  ..?@...H
0x00007FF65C4F1B10  8b c8 ff 15 08 f8 00 00  ??...?..
```

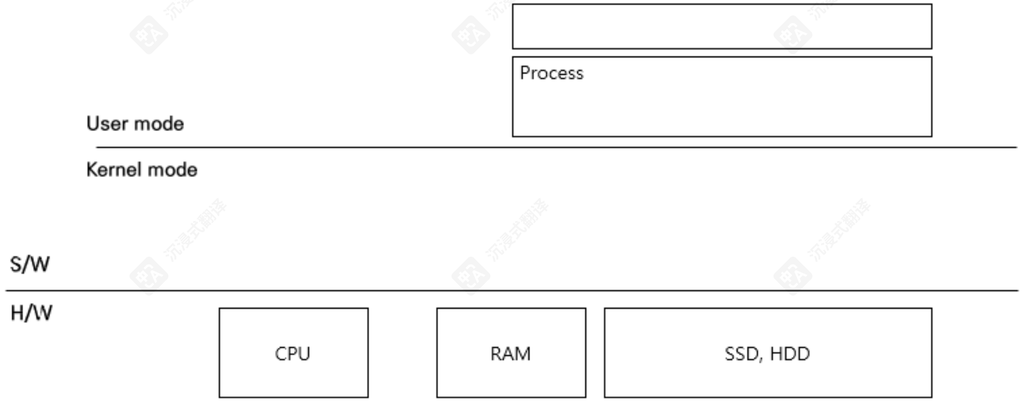
```
mov         qword ptr [rsp+8],rcx
push        rbp
push        rdi
sub         rsp,108h
lea         rbp,[rsp+20h]
lea         rcx,[__AA290256_SingleList@c (
call        __CheckForDebuggerJustMyCode (
```


- 机器码是由电信号（0，1）组成的。
- 它用十六进制表示，而不是二进制。
- **0**x表示十六进制。
- 它通常用反汇编代码表示。

High-level and low-level language

高级语言和低级语言





02

History of Bugs

02

虫的历史

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History of Bugs!!!

Grace Hopper

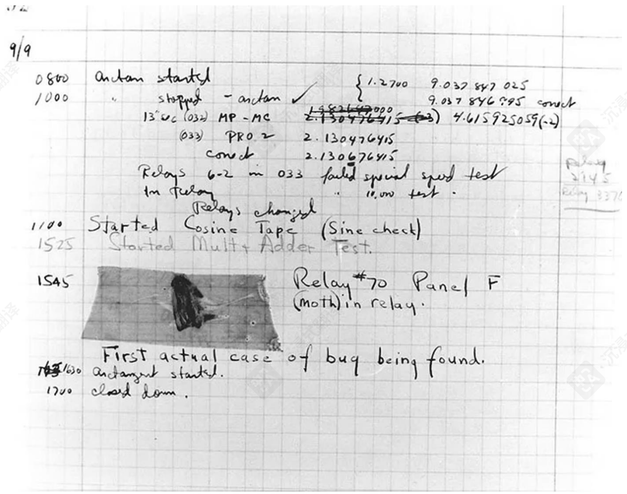
- U.S. Navy Admiral (1906–1992) and computer engineer.
- Participated in computer development projects related to naval gun ballistics calculations.
- Founder of the concept of the compiler.
- Mother of the COBOL(Common Business Oriented Language) programming language.
- NVIDIA H100 (Hopper).

历史!!!

Grace Hopper

- 美国海军上将（1906–1992）和计算机工程师。
- 参与了与海军火炮弹道相关的计算机开发项目
  计算。
- 编译器概念创始人。
- COBOL（通用商业语言）编程语言之母。
- NVIDIA H100 (Hopper)。

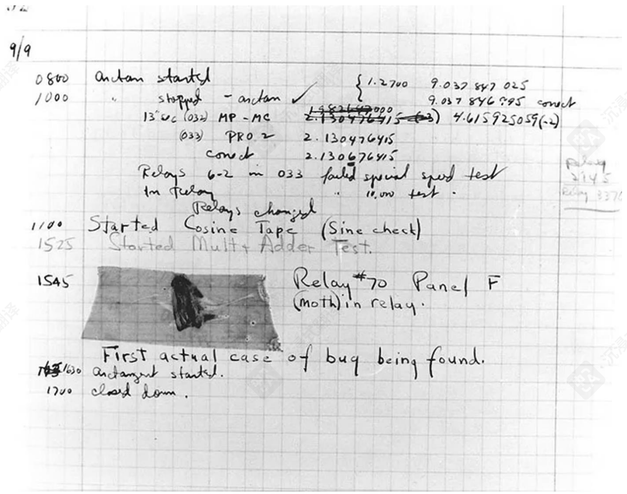
History of Bugs!!!



Grace Hopper

- The cause of the Mark 2 (Harvard) calculation error in 1946 was identified.
- The error was confirmed to be caused by a moth caught in the relay (#70, Panel F), and the findings were recorded.

历史!!!



Grace Hopper

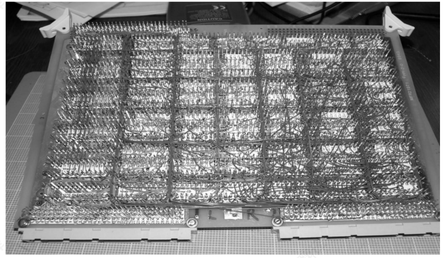
- 1946年Mark 2（哈佛）计算错误的原因被确定。
- 确认错误是由一只飞蛾卡在继电器（#70，面板F）中引起的，并且记录了这一发现。

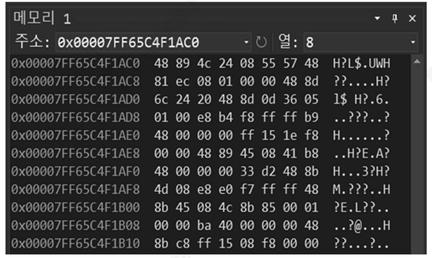
History of Bugs!!!

- COBOL Code

```
IDENTIFICATION DIVISION.
PROGRAM-ID. IDSAMPLE.
ENVIRONMENT DIVISION.
PROCEDURE DIVISION.
    DISPLAY 'HELLO WORLD'.
    STOP RUN.
```

Compile



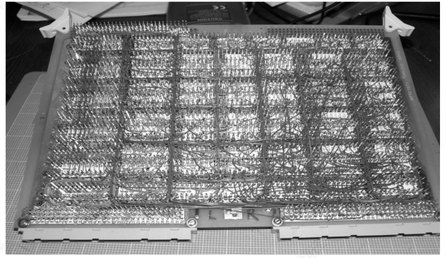


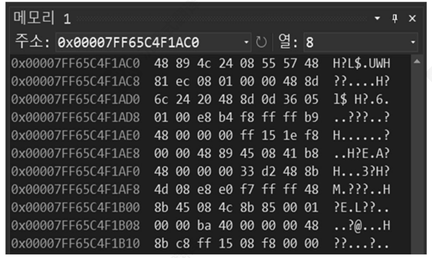
虫的历史!!!

- COBOL代码

```
IDENTIFICATION DIVISION.
PROGRAM-ID. IDSAMPLE.
ENVIRONMENT DIVISION.
PROCEDURE DIVISION.
    DISPLAY 'HELLO WORLD'.
    STOP RUN.
```

编译





03

Terms, Expressions and Statements

03

术语、表达式和语句

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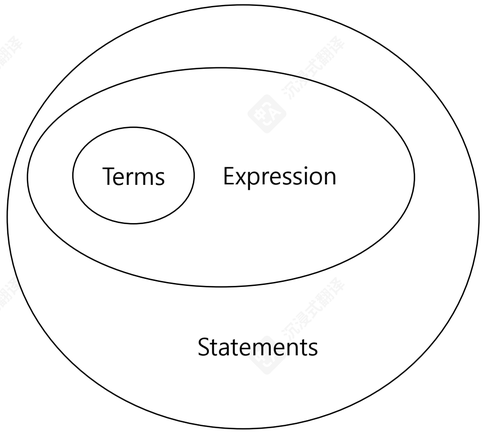
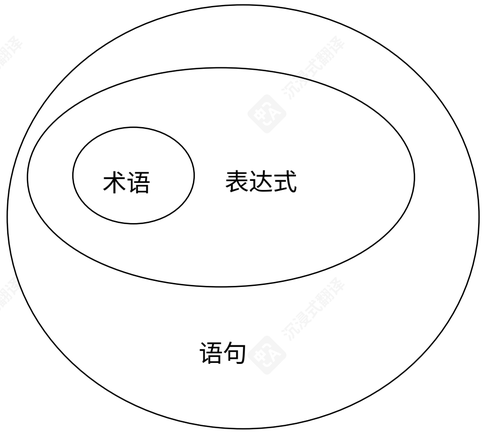
Terms, Expressions, and Statements



- Multiple terms combine to form an expression.
- An expression is the object of evaluation, and **evaluation** requires calculation (operation).
- Multiple equations combine to form a single statement, which serves as the execution unit for high-level language programming.
- **Consecutive statements are executed sequentially from top to bottom.**

术语、表达式和语句



- 多个术语组合为 form an 表达式。
- 表达式是求值对象，求值需要计算(运算)。
- 多个方程组合为单个语句，该语句作为高级语言编程的执行单元。
- **连续语句按从上到下的顺序依次执行。**

Compiler & Interpreter

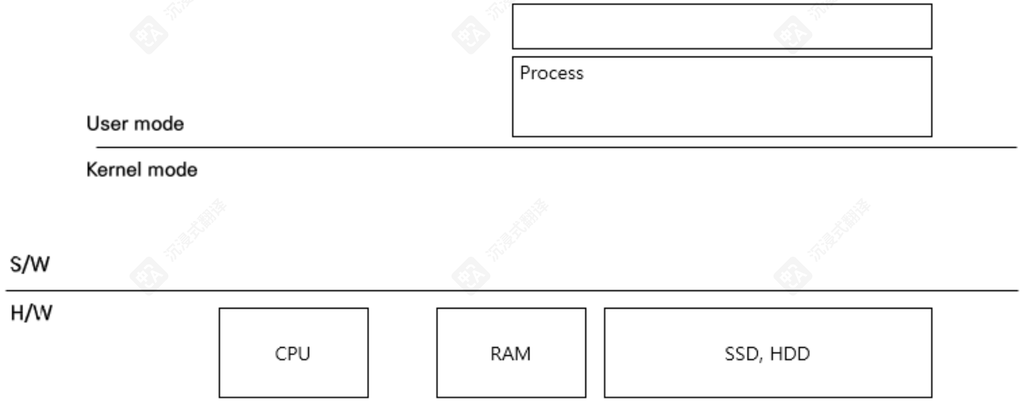
Compiler	Interpreter
• A program that translates source code into machine language. • The entire source code is converted to machine language and then executed. • Adopted by many languages due to its ease of performance **optimization**. • C, C++	• A program or environment that directly executes source code. • Typically executed one line at a time. • Slower than the compiler method in terms of performance (especially speed). • JavaScript (JS) and Python.

编译器 & 解释器

编译器	解释器
• 一个将源代码翻译成机器语言的程序。 • 整个源代码被转换为机器语言然后执行。 • 由于易于性能优化而被许多语言采用。 • C, C++	• 一个直接执行源代码的程序或环境。 • 通常逐行执行。 • 在性能方面比编译器方法慢（尤其是速度）。 • JavaScript (JS) 和 Python。

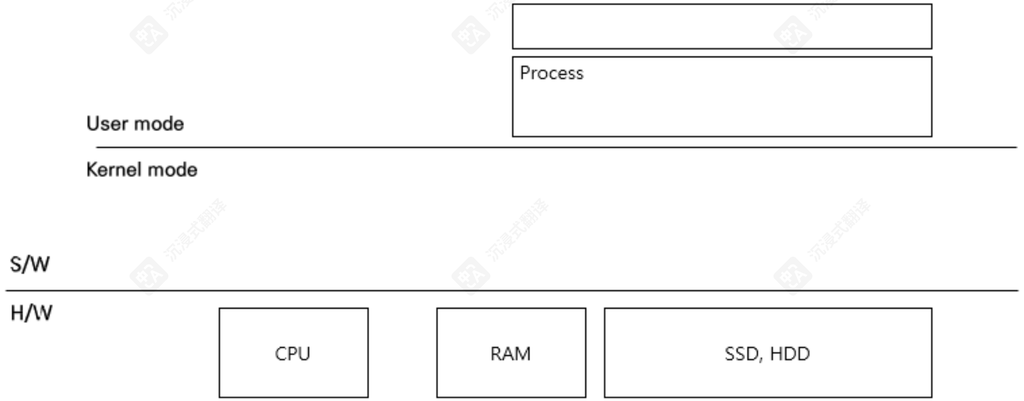
Compiler & Interpreter

JAVA is a hybrid language.
JAVA compiler: **javac.exe**



编译器 & 解释器

JAVA 是一种混合语言。
JAVA 编译器：javac.exe



Quiz

- What are the main characteristics that distinguish high-level and low-level languages?
  A. Syntax complexity
  B. Closeness to hardware
  C. Differences in development speed
  D. Number of supported data types

Quiz

- 高级语言和低级语言的主要区别是什么?

  A. 语法复杂性
  B. 与硬件的接近程度
  C. 开发速度的差异
  D. 支持的数据类型数量

Quiz

- What are the major differences between compilers and interpreters?
  A. When do errors get detected?
  B. How do they translate and execute?
  C. Whether they can be optimized?
  D. What programming paradigms do they support?

Quiz

- 编译器和解释器的主要区别是什么?
  A. 错误何时被检测到?
  B. 它们如何翻译和执行?
  C. 它们是否可以被优化?
  D. 它们支持哪些编程范式?

Quiz

- What is "Bytecode" in Java?
  A. Code directly executed by the CPU
  B. Rules for writing Java source code
  C. Intermediate code understood and executed by the JVM
  D. Code directly converted to native code

Quiz

- Java 中的 "字节码" 是什么?
  A. CPU 直接执行的代码 B. 编写 Java 源代码的规则
  C. 被 JVM 理解和执行的中间代码
  D. 直接转换为本地代码的代码

Quiz

- What are the main reasons why Java is said to be platform-independent?
  A. Because Java code is immediately converted to native code on any operating system.
  B. Because the source code runs directly, unmodified, on any environment.
  C. Because the Java Virtual Machine (JVM) executes bytecode on various operating systems.
  D. Because it uses an interpreter without a compilation process.

Quiz

- Java 被称为平台无关的主要原因是什么?

  A. 因为 Java 代码在任何操作系统上都会立即转换为本地代码。

  B. 因为源代码在任何环境中都能直接、未经修改地运行。
  C. 因为 Java 虚拟机 (JVM) 在各种操作系统上执行字节码。

  D. 因为它使用解释器而没有编译过程。

Quiz

- Among the basic elements of a programming language, what does a statement mean?
  A. piece of code that can be evaluated to a value.
  B. The smallest unit of code used in a word.
  C. A component of code that serves as a unit of program execution.
  D. A named variable or constant.

Quiz

- 编程语言的基本元素中，语句是什么意思?

  A. 可以求值的代码片段。
  B. 字中使用代码的最小单元。
  C. 作为程序执行单元的代码组件。
  D. 命名的变量或常量。

04

What is programming?

04

什么是编程?

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What is programming?

- Programming itself is a **written process** that describes **procedural sequences**.
- It requires training to logically solve problems step by step
  - (a separate issue from programming language grammar).
- Start as simply as possible.

什么是编程?

- 编程本身是一个描述过程性
  序列的书面过程。
- 它需要训练来逐步逻辑地解决问题
  - (这是一个与编程语言语法无关的问题)。
- 尽可能简单开始。

Procedural Writing

程序性写作

Procedural Writing

程序化写作

Procedural Writing

- There are surprisingly many **omitted** expressions in human language.
  1. Bring jam, bread, and a knife and place them on the table.
  2. Use both hands to open the jar containing the jam.
  3. Pick up the knife with your right hand and scoop up a small amount of jam from the jar.
  4. Place the bread in your left hand with the wide side facing up.
  5. Spread the jam attached to the knife evenly over the wide side of the bread.

程序性写作

- 人类语言中竟然有如此多的省略表达。

  1. 带上果酱、面包和一把刀，并将它们放在桌子上。
  2. 用双手打开装有果酱的罐子。
  3. 用右手拿起刀，从罐子里舀起少量果酱。
  4. 用左手拿起面包，让宽面朝上。
  5. 将刀上的果酱均匀地涂抹在面包的宽面上。

Constants and Variables

- Constants
  - A number whose value is determined at **the time the expression is written**.
  - A number whose value is **fixed and cannot change in the future**.
  - A literal constant.
    - ‘A’, “Hello”, 3, 3.4F, 123.45
  - A symbolic constant.
    - final

常量和变量

- 常量
  - 一个在表达式编写时确定值的数字。
  - **一个固定且未来无法改变的数字。**
  - 一个文字常量。
    - ‘A’ ， “你好”, 3, 3.4F, 123.45

  一个符号常量。
    - 最终

Constants and Variables

- Variables
  - A number whose value is **not determined at the time the expression is written**.
  - A number that is **not specified or may change in the future**.
  - The most common way developers **use memory**.
  - A variable that can cause calculations to vary depending on the value specified.
    - Weather variables (clear, cloudy, rain, snow, temperature).
    - The beginning of a dependency.

常量和变量

- 变量
  - **一个在表达式编写时其值未确定的数。**
  - **一个未指定或在将来可能改变的数。**
  - 开发者使用内存的最常见方式。
  - 一个可以导致计算结果根据指定值变化的变量。
    - 天气变量（晴朗、多云、下雨、下雪、温度）。
    - 依赖关系的开始。

Procedural Writing

- Croissants instead of bread?
  1. Bring jam, bread, and a knife and place them on the table.
  2. Use both hands to open the jar containing the jam.
  3. Pick up the knife with your right hand and scoop up a small amount of jam from the jar.
  4. Place the bread in your left hand with the wide side facing up.
  5. Spread the jam attached to the knife evenly over the wide side of the bread.

程序性写作

- 用羊角面包代替面包?
  1. 带上果酱、面包和一把刀，把它们放在桌子上。
  2. 用双手打开装有果酱的罐子。
  3. 用右手拿起刀，从罐子里舀起一小块果酱。
  4. 把面包放在左手，宽面朝上。
  5. 把刀上的果酱均匀地涂抹在面包的宽面上。

Procedural Writing

- Croissants instead of bread?
  1. Bring jam, [redacted], and a knife and place them on the table.
  2. Use both hands to open the jar containing the jam.
  3. Pick up the knife with your right hand and scoop up a small amount of jam from the jar.
  4. Place the [redacted] in your left hand with the wide side facing up.
  5. Spread the jam attached to the knife evenly over the wide side of the [redacted].

程序性写作

- 用羊角面包代替面包?
  1. 带上果酱、面包和一把刀，把它们放在桌子上。
  2. 用双手打开装有果酱的罐子。
  3. 用右手拿起刀，从罐子里舀起一小块果酱。
  4. 把面包放在左手，宽面朝上。
  5. 把刀上的果酱均匀地涂抹在面包的宽面上。

Cases and Flow Control

- Assuming there are only three types of bread:
  - Bread, croissant, and salt bread
- Consider the case where butter is present in addition to jam.
  - Smear only jam
  - Smear only butter
  - Smear both
- Check whether the jam-spreading procedure is still valid.

案例与流程控制

- 假设只有三种面包类型：
  - 面包、羊角面包和咸面包
- 考虑在果酱之外还有黄油的情况。
  - 只涂抹果酱
  - 只涂抹黄油
  - 涂抹两边
- 检查果酱涂抹程序是否仍然有效。

Function

- "Procedure for spreading jam on bread" with reuse in mind

- Procedure for spreading jam on bread( )
  1. Bring jam, [redacted], and a knife and place them on the table.
  2. Use both hands to open the jar containing the jam.
  3. Pick up the knife with your right hand and scoop up a small amount of jam from the jar.
  4. Place the [redacted] in your left hand with the wide side facing up.
  5. Spread the jam attached to the knife evenly over the wide side of the [redacted].

functions, routines, **methods**, procedure

功能

- "在面包上涂抹果酱的步骤"，并考虑重复使用

- 在面包上涂抹果酱的步骤( )
  1. 将果酱、面包和一把刀放在桌子上。
  2. 用双手打开装有果酱的罐子。
  3. 用右手拿起刀，从罐子里舀起少量果酱。
  4. 用左手拿起面包，让宽面朝上。
  5. 将刀上附着的果酱均匀涂抹在面包的宽面上。

函数，过程，方法，程序

Quiz

- What is the most important first step in programming?
  - A. Learning grammar
  - B. Entering code
  - C. Organizing and concretizing thoughts
  - D. Running the program

Quiz

- 编程中最重要的一步是什么?
  - A. 学习语法
  - B. 输入代码
  - C. 组织和具体化想法
  - D. 运行程序

Quiz

- What is the biggest difference between a "variable" and a "constant" in programming?
  A. Memory usage
  B. Types of values that can be stored
  C. When the value is determined
  D. Name length

Quiz

- 变量和常量在编程中的最大区别是什么是什么?
  A. 内存使用
  B. 可以存储的值类型
  C. 值何时确定
  D. 名称长度

Quiz

- What naturally follows from the concept of a variable?
  A. Definition of a function
  B. Cases and flow control
  C. Writing comments
  D. Database connection

Quiz

- 从变量的概念中自然地引申出什么?
  A. 函数的定义
  B. 案例和流程控制
  C. 编写注释
  D. 数据库连接

Quiz

- What is the primary purpose of grouping multiple processing steps into a "function" or "method"?
  A. Reducing program size
  B. Increasing code reusability
  C. Enhancing data security
  D. Automatic error message generation

Quiz

- 将多个处理步骤组合成一个 "函数" 或 "方法" 的主要目的是什么?

  A. 减小程序大小
  B. 提高代码可重用性
  C. 增强数据安全性
  D. 自动生成错误消息

Quiz

- What are the values that can be determined or changed during program execution?
  A. Literal constants
  B. Symbolic constants
  C. Variables
  D. Function names

Quiz

- 在程序执行期间可以确定或更改哪些值?

  A. 字面常量
  B. 符号常量
  C. 变量
  D. 函数名

Q & A

问答

THANK YOU

谢谢